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7.23 $p(t)=\sum_{k=-\infty}^{+\infty}(-1)^k\delta(t-k\Delta) \quad = \sum_{k=-\infty}^{+\infty}\frac{(-1)^k}{\Delta}e^{-jk(\frac{2\pi}{\Delta})t}$

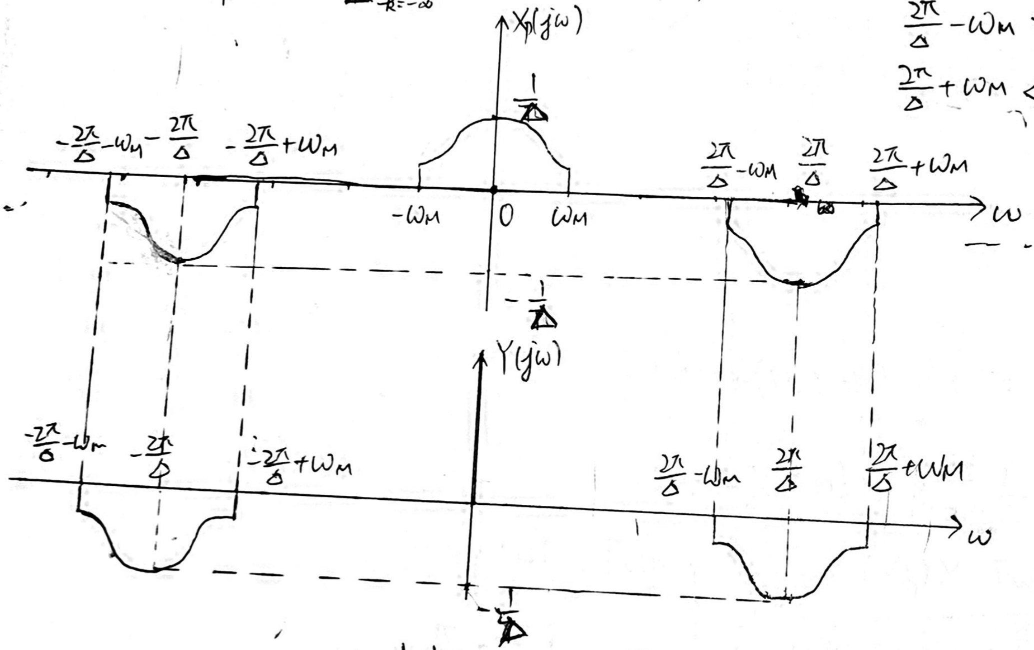
$\Leftrightarrow P(j\omega)=\frac{2\pi}{\Delta}\sum_{k=-\infty}^{+\infty}(-1)^k\delta(\omega-\frac{2\pi k}{\Delta})$

$\frac{2\pi}{\Delta} > \frac{2\pi}{\pi}\cdot 2\omega_M \Rightarrow 4\omega_M$

$\frac{2\pi}{\Delta}-\omega_M > \frac{2\pi}{\Delta}-\frac{\pi}{2\Delta}=\frac{3}{2}\cdot\frac{\pi}{\Delta} > \frac{\pi}{\Delta} > \omega_M$

$\frac{2\pi}{\Delta}+\omega_M < \frac{2\pi}{\Delta}+\frac{\pi}{2\Delta}=\frac{5}{2}\cdot\frac{\pi}{\Delta} < \frac{3\pi}{\Delta}$

(a) $X_p(j\omega)=\frac{1}{\Delta}\sum_{k=-\infty}^{+\infty}(-1)^k X(j(\omega-\frac{2\pi k}{\Delta}))$



(b) System $H(j\omega)=\begin{cases}1, & |\omega|<\omega_c, \quad \omega_c\in[\omega_M, \frac{2\pi}{\Delta}-\omega_M] \\ 0, & \text{otherwise.}\end{cases}$

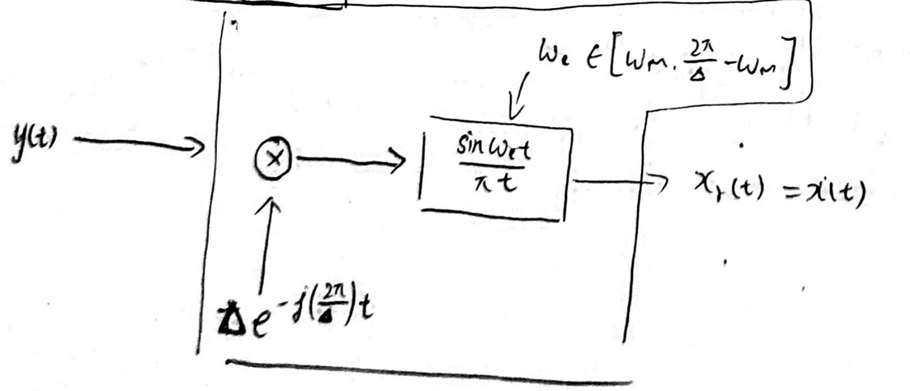
$h_r(t)=\frac{1}{2\pi}\int_{-\infty}^{+\infty}H(j\omega)e^{j\omega t}d\omega=\frac{1}{2\pi}\int_{-\omega_c}^{\omega_c}e^{j\omega t}d\omega=\frac{1}{2\pi jt}(e^{j\omega_c t}-e^{-j\omega_c t})$

$=\frac{1}{\pi t}\cdot\sin\omega_c t=\frac{\sin\omega_c t}{\pi t}$

$x_p(t) \longrightarrow \boxed{h_r(t)} \longrightarrow x_r(t)=x(t)$

~~(c) System:~~

(c)



(d) the maximum value of $\Delta = \Delta_{max}$ satisfies.

$$\frac{2\pi}{\Delta_{max}} - \omega_M = \omega_M \Leftrightarrow \Delta_{max} = \frac{\pi}{\omega_M}$$

7.36 (a) $p(t) = \sum_{k=-\infty}^{+\infty} \delta(t-kT)$

$$\frac{dx(t)}{dt} = \sum_{n=-\infty}^{+\infty} x(nT)\, g(t-nT) = (x(t)p(t)) * g(t)$$

$$j\omega X(j\omega) = G(j\omega)\cdot\frac{1}{T}\sum_{k=-\infty}^{+\infty} X\left(j\left(\omega-\frac{2\pi k}{T}\right)\right) \Leftarrow G(j\omega) = \begin{cases} 2\pi j, & |\omega|<\frac{\pi}{T} \\ 0, & |\omega| \geq \frac{\pi}{T}\end{cases}$$

$$\cancel{G(j\omega) = \begin{cases} 2\pi j \end{cases}}$$

$$g(t) = \frac{1}{2\pi}\int_{-\frac{\pi}{T}}^{\frac{\pi}{T}} 2\pi j\, e^{j\omega t}\, d\omega = \frac{1}{t}\left(e^{jt\cdot\frac{\pi}{T}} - e^{jt\cdot(-\frac{\pi}{T})}\right)$$

$$= \frac{2j}{t}\sin\left(\frac{\pi}{T}\cdot t\right)$$

(b)

7.36 (a) $p(t) = \sum_{k=-\infty}^{+\infty} \delta(t-kT)$

$$\frac{dx(t)}{dt} = \sum_{n=-\infty}^{+\infty} x(nT)\, g(t-nT) = (x(t)p(t)) * g(t)$$

$$j\omega X(j\omega) = G(j\omega)\cdot\frac{1}{T}\sum_{n=-\infty}^{+\infty} X\left(j\left(\omega-\frac{2\pi k}{T}\right)\right)$$

$$G(j\omega) = \frac{j\omega T\, X(j\omega)}{\sum_{n=-\infty}^{+\infty} X\left(j\left(\omega-\frac{2\pi n}{T}\right)\right)} = \begin{cases} j\omega T, & |\omega| < \frac{\pi}{T} \\ 0, & |\omega| \geq \frac{\pi}{T}\end{cases}$$

$$g(t) = \frac{1}{2\pi}\int_{-\frac{\pi}{T}}^{\frac{\pi}{T}} j\omega T\, e^{j\omega t}\, d\omega = \frac{T}{2\pi t}\left[\omega e^{j\omega t}\Big|_{-\frac{\pi}{T}}^{\frac{\pi}{T}} - \int_{-\frac{\pi}{T}}^{\frac{\pi}{T}} e^{j\omega t}\, d\omega\right]$$

$$= \frac{T}{2\pi t}\left(\frac{\pi}{T}e^{j(\frac{\pi}{T})t} + \frac{\pi}{T}e^{-j(\frac{\pi}{T})t} - \frac{1}{jt}\left(e^{j(\frac{\pi}{T})t} - e^{j(-\frac{\pi}{T})t}\right)\right)$$

$$= \frac{T}{2\pi t}\left(\frac{2\pi}{T}\cos\left(\frac{\pi}{T}t\right) - \frac{2}{t}\sin\left(\frac{\pi}{T}t\right)\right) = \frac{\cos(\frac{\pi}{T}t)}{t} - \frac{T}{\pi t^2}\sin\left(\frac{\pi}{T}t\right)$$

(b) 是

7.37 (a) $p(t) = \sum_{n=-\infty}^{+\infty}\delta\left(t+\frac{2\pi n}{W}\right) + \sum_{n=-\infty}^{+\infty}\delta\left(t-\Delta-\frac{2\pi n}{W}\right) = \sum_{n=-\infty}^{+\infty}\left(\delta\left(t-\frac{2\pi n}{W}\right) + \delta\left(t-\Delta-\frac{2\pi n}{W}\right)\right)$

$$P(j\omega) = \frac{2\pi}{\frac{2\pi}{W}}\sum_{n=-\infty}^{+\infty}\delta(\omega-nW)\cdot(1+e^{-j\omega\Delta}) = W(1+e^{-j\omega\Delta})\sum_{n=-\infty}^{+\infty}\delta(\omega-nW)$$

清華大學

$$y_1(t) = x(t)\,p(t)\,f(t) = \left[\sum_{n=-\infty}^{+\infty} x\left(\frac{2\pi n}{W}\right)\delta\left(t-\frac{2\pi n}{W}\right) + \sum_{n=-\infty}^{+\infty} x\left(\Delta+\frac{2\pi n}{W}\right)\delta\left(t-\Delta-\frac{2\pi n}{W}\right)\right] f(t)$$

$$= a\sum_{n=-\infty}^{+\infty} x\left(\frac{2\pi n}{W}\right)\delta\left(t-\frac{2\pi n}{W}\right) + b\sum_{n=-\infty}^{+\infty} x\left(\Delta+\frac{2\pi n}{W}\right)\delta\left(t-\Delta-\frac{2\pi n}{W}\right)$$

$$\cancel{Y_1(j\omega) = a}$$

$$Y_1(j\omega) = a\cdot\frac{W}{2\pi}\sum_{n=-\infty}^{+\infty} X(\omega-nW) + b\cdot\frac{1}{2\pi}\int_{-\infty}^{+\infty} X(j\theta)\sum_{n=-\infty}^{+\infty} e^{-j(\omega-\theta)\Delta}\cdot\frac{W}{2\pi}\,\delta(\omega-\theta-nW)\,d\theta$$

$$= \frac{aW}{2\pi}\sum_{n=-\infty}^{+\infty} X(j(\omega-nW)) + \frac{bW}{(2\pi)^2}\sum_{n=-\infty}^{+\infty} X(j(\omega-nW))\,e^{-jnW\Delta}$$

$$= \frac{W}{2\pi}\left[a\sum_{n=-\infty}^{+\infty} X(j(\omega-nW)) + \frac{b}{2\pi}\sum_{n=-\infty}^{+\infty} e^{-jnW\Delta} X(j(\omega-nW))\right]$$

$$Y_2(j\omega) = Y_1(j\omega)\,H_1(j\omega) = \frac{\mathrm{sgn}(\omega)}{2\pi}Wj\left[a\sum_{n=-\infty}^{+\infty} X(j(\omega-nW)) + \frac{b}{2\pi}\sum_{n=-\infty}^{+\infty} e^{-jnW\Delta} X(j(\omega-nW))\right]$$

$$\cancel{Y_3(j\omega) = \frac{W}{2\pi}\left(1+e^{-jnW\Delta}\right)\sum_{n=-\infty}^{+\infty} X(j(\omega-nW))}$$

$$Y_3(j\omega) = \frac{W}{2\pi}\left[\sum_{n=-\infty}^{+\infty} X(j(\omega-nW)) + \frac{1}{2\pi}\sum_{n=-\infty}^{+\infty} e^{-jnW\Delta} X(j(\omega-nW))\right]$$

(b) $Y_2(j\omega) + Y_3(j\omega) = \dfrac{X(j\omega)}{H_2(j\omega)}$ ，考虑 $0<\omega<W$

$$\cancel{\frac{W}{2\pi}\sum_{n=-\infty}^{+\infty}\left(\mathrm{sgn}(\omega)\,ja+1\right.}$$

$$\cancel{Y_2(j\omega)+Y_3(j\omega) = \frac{W}{2\pi}\left(ja+1+\frac{b+1}{2\pi}\right.}$$

$$\cancel{Y_2(j\omega)+Y_3(j\omega) = \frac{W}{2\pi}\sum_{n=-\infty}^{+\infty}\left(ja+\frac{b}{2\pi}je^{-jnW\Delta}+1+\frac{1}{2\pi}e^{-jnW\Delta}\right)X(j(\omega-nW))}$$

$$\cancel{H_2(j\omega)\left(Y_2(j\omega)+Y_3(j\omega)\right) =}$$

$$Y_2(j\omega)+Y_3(j\omega) = \frac{W}{2\pi}\sum_{n=-\infty}^{+\infty}\left(\mathrm{sgn}(\omega)\,j\left(a+\frac{b}{2\pi}e^{-jnW\Delta}\right)+1+\frac{1}{2\pi}e^{-jnW\Delta}\right)X(j(\omega-nW))$$

考虑 (i) $0<\omega<W$

$$H_2(j\omega)\left(Y_2(j\omega)+Y_3(j\omega)\right)=K\left[\left(j\left(a+\frac{b}{2\pi}\right)+\left(1+\frac{1}{2\pi}\right)\right)X(j\omega)+\left(j\left(a+\frac{b}{2\pi}e^{-jW\Delta}\right)+1+\frac{1}{2\pi}e^{-jW\Delta}\right)X(j(\omega-W))\right]$$

(ii) $-W<\omega<0$

$$H_2(j\omega)\left(Y_2(j\omega)+Y_3(j\omega)\right)=K^*\left[\left(-j\left(a+\frac{b}{2\pi}\right)+\left(1+\frac{1}{2\pi}\right)\right)X(j\omega)+\left(-j\left(a+\frac{b}{2\pi}e^{jW\Delta}\right)+1+\frac{1}{2\pi}e^{jW\Delta}\right)X(j(\omega+W))\right]$$

$$\begin{cases} K\left[\left(1+\frac{1}{2\pi}\right)+j\left(a+\frac{b}{2\pi}\right)\right]=1 \\ \left[\left(1+\frac{1}{2\pi}\cos W\Delta+\frac{b}{2\pi}\sin W\Delta\right)+j\left(\frac{1}{2\pi}\sin W\Delta-\frac{b}{2\pi}\cos W\Delta-a\right)\right]=0 \end{cases}$$

$$\Leftrightarrow\ K=\frac{1}{\left(1+\frac{1}{2\pi}\right)+j\left(a+\frac{b}{2\pi}\right)} \qquad \cancel{\begin{bmatrix}\cos W\Delta & \sin W\Delta \\ -\sin W\Delta & \cos W\Delta\end{bmatrix}\begin{bmatrix}\frac{1}{2\pi} \\ \frac{b}{2\pi}\end{bmatrix}=\begin{bmatrix}-1 \\ -a\end{bmatrix}}$$

$$b=-\frac{1+\frac{1}{2\pi}\cos W\Delta}{\frac{1}{2\pi}\sin W\Delta}=-\frac{2\pi+\cos W\Delta}{\sin W\Delta}=-\frac{2\pi}{\sin W\Delta}-\cot W\Delta$$

$$a=\frac{1}{2\pi}\left(\sin W\Delta+\frac{2\pi+\cos W\Delta}{\sin W\Delta}\cos W\Delta\right)=\frac{1}{2\pi}\left(\frac{(2\pi+1)\cos W\Delta}{\sin W\Delta}\right)=\left(1+\frac{1}{2\pi}\right)\cot W\Delta$$

$$K=\frac{1}{\left(1+\frac{1}{2\pi}\right)+j\left(\cot W\Delta+\cancel{\frac{1}{2\pi}\cot W\Delta}-\frac{1}{\sin W\Delta}-\cancel{\frac{1}{2\pi}\cot W\Delta}\right)}$$

$$=\frac{1}{\left(1+\frac{1}{2\pi}\right)-j\tan\frac{W\Delta}{2}}=\cancel{\frac{\left(1+\frac{1}{2\pi}\right)+j\tan\frac{W\Delta}{2}}{\sqrt{\left(1+\frac{1}{2\pi}\right)^2+\tan^2\frac{W\Delta}{2}}}}$$